



**BERGISCHE
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Master Thesis

Static Quark–Antiquark Sources from Laplacian Eigenmodes and Gluonic Flux-Tube Profiles in Lattice QCD

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Wuppertal, Germany

Abstract

This thesis studies the construction and numerical validation of static quark–antiquark trial states based on low-lying eigenmodes of the three-dimensional gauge-covariant lattice Laplace operator. The central long-term goal is to use these sources to measure gluonic flux-tube profiles through local action-density and energy-density insertions in lattice QCD.

The first stage validates the Laplacian-eigenmode source construction by computing static correlators $L(R, \tau)$, extracting effective energies V_{eff} , and benchmarking the resulting static potential against the expected Cornell form. The present implementation uses static perambulators built from Laplacian eigenvectors and temporal Wilson lines, averages over several temporal source positions, and applies correlated plateau and potential fits. The current numerical status is a validated production pipeline for $R/a = 1, \dots, 6$ with a stable plateau window around $\tau/a = 3, 4, 5$, together with diagnostic scans at larger separations.

The second stage, to be completed as the main physics result of the thesis, introduces local gluonic probes based on plaquette or clover estimators of E^2 and B^2 . The normalized, vacuum-subtracted ratio of the source-probe correlator to the source correlator will then provide the spatial modification of the gluonic action density induced by the static $Q\bar{Q}$ pair.

Zusammenfassung

TODO / thesis-writing note

Write a German summary after the English abstract has stabilized. Keep the physics content parallel to the abstract, not merely a word-by-word translation.

Acknowledgements

TODO / thesis-writing note

Add personal acknowledgements near the end of the thesis. Mention both advisors, the research group, collaborators who helped with the QCD library or numerical setup, and computing resources such as Stromboli/Laplace if appropriate.

Declaration

I hereby declare that I wrote this thesis independently and that I used only the cited sources and aids.

Wuppertal, <date>

Dhruv Patel

Contents

Abstract	iii
Zusammenfassung	v
Acknowledgements	vii
Declaration	ix
Abbreviations and notation	xvii
1. Introduction	1
1.1. Project aim	1
1.2. Research questions	1
1.3. Structure of the thesis	2
2. Theoretical background	3
2.1. Static sources and Wilson loops	3
2.2. Flux tubes and local gluonic probes	3
2.3. Gauge-covariant Laplacian and distillation	4
3. Laplacian-eigenmode static sources	5
3.1. Laplace trial state	5
3.2. Static perambulators	5
3.3. Profile choices and later extensions	5
4. Numerical setup and implementation	7
4.1. Code path	7
4.2. Current production parameters	7
4.3. Statistical analysis	7
5. Static-potential benchmark	9
5.1. Effective-energy plateaus	9
5.2. Cornell fit	9
5.3. Extended diagnostic scan	9
6. Local gluonic probes and flux-tube profiles	13
6.1. Field-strength estimators	13
6.2. Normalized vacuum-subtracted observable	13
6.3. Planned presentation of final results	13
7. Statistical and systematic uncertainties	15
7.1. Fit-window dependence	15
7.2. Temporal-source averaging	15
7.3. Autocorrelations	15
7.4. Eigenvector truncation and source overlap	15

7.5. Probe definition	15
8. Conclusions and outlook	17
A. Reproducibility notes	19
A.1. Suggested repository layout	19
A.2. Compile commands	19
A.3. Suggested Git ignore entries	19
B. Additional diagnostic plots	21

List of Figures

5.1.	<i>Preliminary</i> correlated plateau analysis for $R/a = 1, \dots, 6$ using temporal-source averaging. The highlighted window $\tau/a = 3, 4, 5$ is used for the benchmark static-potential extraction.	9
5.2.	<i>Preliminary</i> correlated Cornell fit to the validated $R/a = 1, \dots, 6$ static-potential points.	10
5.3.	<i>Preliminary</i> fit-stability diagnostic for the Cornell benchmark. This plot is retained in the thesis draft to document how the chosen fit range was assessed.	10
5.4.	<i>Preliminary</i> reliability diagnostics for the extended scan. The maps are used to define safe parameter windows before the flux-tube probe measurements are performed.	11
5.5.	<i>Preliminary</i> diagnostic extension of the static potential beyond the validated range using a common $\tau/a = 3$ estimate. The larger- R points are useful for trend checks but are not yet final precision data.	11
B.1.	<i>Preliminary</i> plateau-window comparison for selected larger-distance diagnostic cases.	21

List of Tables

4.1. Current production and diagnostic parameters for the Laplace-static bench-
mark. 7

Abbreviations and notation

Symbol / abbreviation	Meaning
QCD	Quantum Chromodynamics
LQCD	Lattice Quantum Chromodynamics
$Q\bar{Q}$	Static quark–antiquark pair
R or r	Spatial separation of the static sources
T or τ	Euclidean temporal extent of a static correlator
N_v	Number of Laplacian eigenvectors retained
$L(R, \tau)$	Laplace-static source correlator
$V_{\text{eff}}(R, \tau)$	Effective energy from a correlator ratio
GEVP	Generalized eigenvalue problem
HYP / APE	Common gauge-link smearing schemes

1. Introduction

Quantum Chromodynamics (QCD) is the gauge theory of the strong interaction. Its non-perturbative low-energy regime contains phenomena such as confinement, hadron formation, and the emergence of a chromoelectric flux tube between static color sources. Lattice QCD provides a first-principles framework in which these phenomena can be investigated numerically by discretizing Euclidean space-time and evaluating path integrals with Monte Carlo methods.

A central observable in this setting is the static quark–antiquark potential $V(R)$. Traditionally it is extracted from Wilson loops, where spatial Wilson lines connect the static sources and temporal Wilson lines propagate them in Euclidean time. For detailed studies of flux tubes, string breaking, or momentum-space static potentials, many source separations are desirable, including off-axis separations. This motivates alternative source constructions that preserve gauge covariance while reducing the cost of building spatial transporters.

This thesis follows the idea of replacing spatial Wilson lines by outer products of eigenvectors of the three-dimensional gauge-covariant Laplace operator. The construction is closely related to distillation, where low-lying Laplacian eigenmodes define a smooth low-rank subspace for hadronic operators [1]. For static $Q\bar{Q}$ sources, Laplacian eigenmode trial states and static perambulators provide a flexible way to construct correlators for many source separations [2].

1.1. Project aim

The project aim is two-stage:

1. construct and validate Laplacian-eigenmode static $Q\bar{Q}$ sources through the static potential;
2. use the validated source correlator as the normalization for local gluonic probes, in order to map the flux-tube profile around the static pair.

The current production stage has completed the first validation at short to moderate separations and is entering the local-probe stage.

1.2. Research questions

The thesis is organized around the following questions:

1. Can the implemented Laplace-static correlator reproduce stable effective-energy plateaus and a physically sensible static potential?
2. Which separations and Euclidean time windows are reliable with the present statistics and eigenmode profile choice?
3. How should local action-density and energy-density insertions be normalized and vacuum-subtracted so that they measure the gluonic response of the gauge field to a static $Q\bar{Q}$ pair?

4. What systematic effects arise from the number of eigenvectors, temporal-source averaging, fit-window selection, and autocorrelations?

1.3. Structure of the thesis

Chapter 2 reviews the relevant lattice-QCD background. Chapter 3 derives the Laplacian-eigenmode trial states and static correlators. Chapter 4 documents the implementation and current production setup. Chapter 5 presents the static-potential benchmark. Chapter 6 develops the local-probe formalism for the final flux-tube measurements. Chapter 7 discusses statistical and systematic uncertainties, and Chapter 8 summarizes the current conclusions and outlook.

2. Theoretical background

2.1. Static sources and Wilson loops

For a static color source and antsource separated by $R = |\mathbf{y} - \mathbf{x}|$, the conventional trial state contains a spatial Wilson line $U_s(\mathbf{x}, \mathbf{y})$ connecting the two positions. Under a gauge transformation G , it transforms as

$$U_s(\mathbf{x}, \mathbf{y}) \longrightarrow G(\mathbf{x})U_s(\mathbf{x}, \mathbf{y})G^\dagger(\mathbf{y}), \quad (2.1)$$

so that the full static meson operator is gauge invariant. The corresponding rectangular Wilson loop behaves at large Euclidean time as

$$W(R, T) \propto \exp[-V(R)T] \quad (2.2)$$

up to excited-state contributions and normalization factors. The effective energy can therefore be estimated from a ratio,

$$V_{\text{eff}}(R, T) = \log \frac{W(R, T)}{W(R, T + a)}. \quad (2.3)$$

At intermediate distances, the static potential is commonly parametrized by a Cornell form,

$$aV(R) = aV_0 + \sigma a^2 R - \frac{\alpha}{R}, \quad (2.4)$$

where σ is the string tension in lattice units and α parametrizes the Coulombic contribution.

2.2. Flux tubes and local gluonic probes

The formation of a flux tube can be probed by inserting local field-strength observables into the source correlator. In terms of Euclidean electric and magnetic components one may define an action-density probe

$$S(\mathbf{z}, t) = \frac{1}{2} \left[E^2(\mathbf{z}, t) + B^2(\mathbf{z}, t) \right], \quad (2.5)$$

and an energy-density probe

$$\epsilon(\mathbf{z}, t) = \frac{1}{2} \left[E^2(\mathbf{z}, t) - B^2(\mathbf{z}, t) \right]. \quad (2.6)$$

The observable of interest is not the raw expectation value, but the source-induced modification after vacuum subtraction. For an optimized source correlator $C_{Q\bar{Q}}$, the schematic ratio is

$$\Delta S_{Q\bar{Q}}(\mathbf{z}; R, \tau) = \frac{\langle C_{Q\bar{Q}}(R, \tau) S(\mathbf{z}, t_{\text{mid}}) \rangle}{\langle C_{Q\bar{Q}}(R, \tau) \rangle} - \langle S(\mathbf{z}, t_{\text{mid}}) \rangle. \quad (2.7)$$

This construction follows the logic used in previous lattice studies of action-density profiles and string breaking [3].

2.3. Gauge-covariant Laplacian and distillation

On each time slice, the three-dimensional gauge-covariant Laplacian acts on a color vector field as

$$\Delta\psi(\mathbf{x}) = \frac{1}{a^2} \sum_{k=1}^3 \left[U_k^\dagger(\mathbf{x} - a\hat{k})\psi(\mathbf{x} - a\hat{k}) - 2\psi(\mathbf{x}) + U_k(\mathbf{x})\psi(\mathbf{x} + a\hat{k}) \right]. \quad (2.8)$$

The eigenvalues are gauge invariant, while the eigenvectors transform covariantly. Therefore outer products of eigenvectors can be arranged to transform like spatial Wilson lines. Distillation uses a low-rank projector onto the subspace spanned by the lowest N_v eigenvectors,

$$\square(\mathbf{z}, \mathbf{x}; t) = \sum_{i=1}^{N_v} v_i(\mathbf{z}, t) v_i^\dagger(\mathbf{x}, t), \quad (2.9)$$

which suppresses ultraviolet modes and gives a flexible operator basis [1].

3. Laplacian-eigenmode static sources

3.1. Laplace trial state

The Laplace-static construction replaces the spatial Wilson line by a sum of Laplacian eigenvector outer products. With optional eigenmode weights ρ_i , the source operator between $\mathbf{x}$ and $\mathbf{y}$ is built from

$$\Phi(\mathbf{x}, \mathbf{y}; t) = \bar{Q}(\mathbf{x}, t) \left[\sum_{i=1}^{N_v} \rho_i^2 v_i(\mathbf{x}, t) v_i^\dagger(\mathbf{y}, t) \right] Q(\mathbf{y}, t). \quad (3.1)$$

The finite truncation in N_v , or a non-trivial profile ρ_i , is essential: if the complete eigenbasis is used with trivial weights, the source collapses to a local identity rather than an extended $Q\bar{Q}$ operator.

3.2. Static perambulators

For a static color source at spatial position $\mathbf{x}$, the temporal Wilson line from source time t_0 to sink time t_1 is projected into the Laplacian subspace through the static perambulator

$$\tau_{ij}(\mathbf{x}; t_0, t_1) = v_i^\dagger(\mathbf{x}, t_0) U_t(\mathbf{x}; t_0, t_1) v_j(\mathbf{x}, t_1). \quad (3.2)$$

The resulting Laplace-static correlator is

$$L(R, \tau) = \left\langle \sum_{i,j=1}^{N_v} \rho_i^2(t_0) \rho_j^2(t_1) \tau_{ij}(\mathbf{y}; t_0, t_1) \tau_{ji}(\mathbf{x}; t_1, t_0) \right\rangle, \quad (3.3)$$

with $R = |\mathbf{y} - \mathbf{x}|$ and $\tau = t_1 - t_0$. The static potential is then extracted from

$$V_{\text{eff}}(R, \tau) = \log \frac{L(R, \tau)}{L(R, \tau + a)}. \quad (3.4)$$

3.3. Profile choices and later extensions

The current production tests use $N_v = 10$ with constant weights $\rho_i = 1$. This is intentionally simple: it validates the data flow and correlator normalization before introducing optimized Gaussian profiles or a GEVP basis. The literature indicates that larger eigenvector bases and Gaussian profile functions can improve ground-state overlap and lead to earlier plateaus [2].

TODO / thesis-writing note

Later update this chapter with the final profile strategy used in the thesis: constant profile only, larger N_v , Gaussian profile basis, or GEVP-optimized profiles.

4. Numerical setup and implementation

4.1. Code path

The implementation is organized around the program `src/test_laplace_RT_scan.c` and the accompanying analysis scripts in `laplace_static/scripts`. The production code path is:

1. read gauge fields and precomputed Laplacian eigenvectors from the existing QCD-library/HDF5 setup;
2. build static perambulators $\tau_{ij}(\mathbf{x}; t_0, t_1)$ for the selected time extent τ ;
3. contract pairs of static perambulators to form $L(R, \tau)$;
4. average over spatial positions, equivalent orientations, configurations, and selected temporal source positions;
5. analyze effective energies, plateau windows, potential fits, and reliability diagnostics.

4.2. Current production parameters

The current benchmark setup is summarized in Table 4.1. These values reflect the validated production stage at which the thesis write-up is being started.

Table 4.1.: Current production and diagnostic parameters for the Laplace-static benchmark.

Quantity	Current value
Gauge configurations	$N_{\text{cfg}} = 100$
Eigenvectors	$N_v = 10$
Eigenmode profile	constant, $\rho_i = 1$
Temporal-source average	$t_0/a = 0, 8, 16, 24, 32, 40$
Validated scan	$R/a = 1, \dots, 6, \tau/a = 1, \dots, 8$
Diagnostic scan	$R/a = 1, \dots, 12, \tau/a = 1, \dots, 10$
Stable benchmark window	$\tau/a = 3, 4, 5$

4.3. Statistical analysis

The analysis uses correlated fits where possible, with jackknife and Gamma-method style cross-checks. The Gamma method estimates autocorrelation effects through explicit autocorrelation functions and an automatically chosen summation window [4]. For derived observables and fit quantities, the thesis should distinguish clearly between the central estimator, the covariance estimate, the fit-window choice, and the treatment of autocorrelations.

TODO / thesis-writing note

Once the final analysis scripts are frozen, document exact commands, file names, software versions, and random/fit settings in this chapter and in Chapter A.

5. Static-potential benchmark

5.1. Effective-energy plateaus

The first validation target is the presence of stable effective-energy plateaus. For $R/a = 1, \dots, 6$, a common fit window $\tau/a = 3, 4, 5$ gives acceptable correlated constant fits. Figure 5.1 shows the corresponding preliminary plateau analysis.

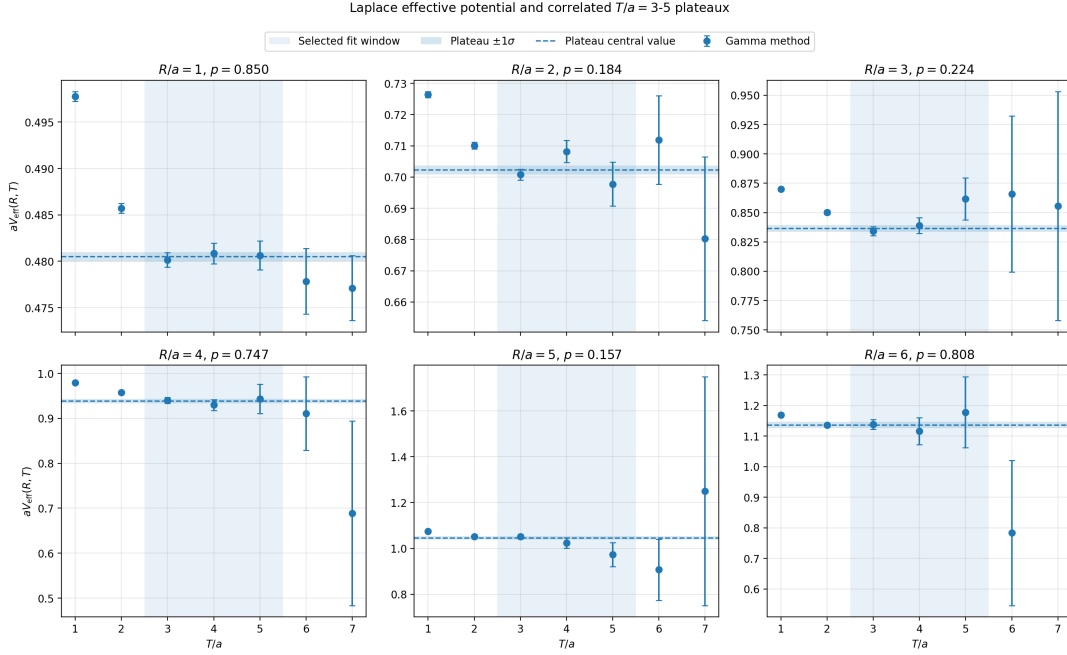


Figure 5.1.: *Preliminary* correlated plateau analysis for $R/a = 1, \dots, 6$ using temporal-source averaging. The highlighted window $\tau/a = 3, 4, 5$ is used for the benchmark static-potential extraction.

5.2. Cornell fit

The plateau values are fitted to the Cornell form in Eq. (2.4). The current benchmark fit gives

$$aV_0 = 0.6646(82), \quad \sigma a^2 = 0.08646(24), \quad \alpha = 0.2705(60), \quad (5.1)$$

with $a\sqrt{\sigma} = 0.2940(40)$, $\chi^2/\text{dof} \approx 1.11$, and $p = 0.344$. These values should be treated as validation-stage results rather than the final physics output.

5.3. Extended diagnostic scan

The extended scan probes $R/a \leq 12$ and $\tau/a \leq 10$. It is not yet used as a final potential extraction at all separations. Instead, it classifies safe, marginal, and diagnostic regions

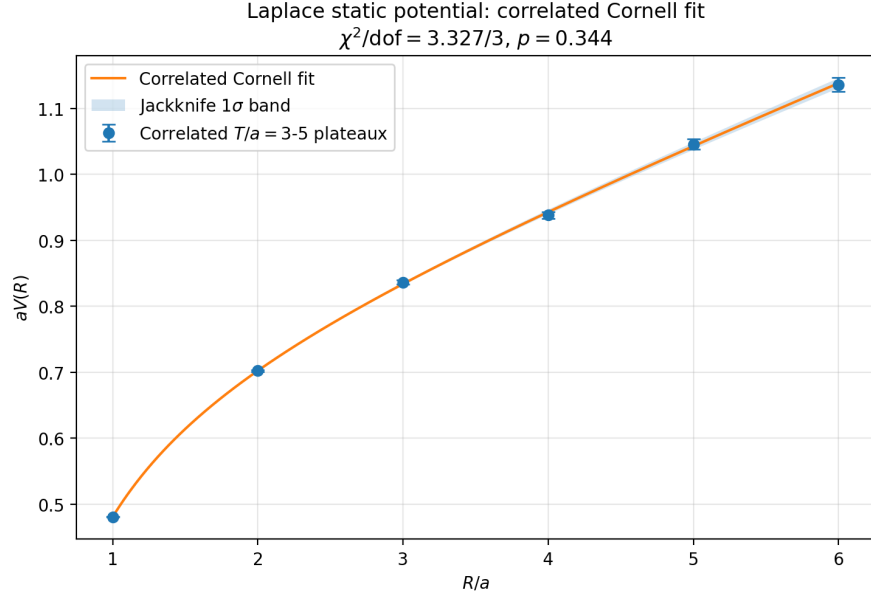


Figure 5.2.: *Preliminary* correlated Cornell fit to the validated $R/a = 1, \dots, 6$ static-potential points.

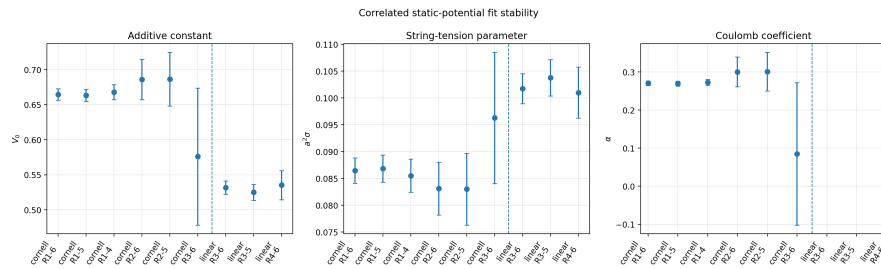


Figure 5.3.: *Preliminary* fit-stability diagnostic for the Cornell benchmark. This plot is retained in the thesis draft to document how the chosen fit range was assessed.

based on relative uncertainty and sign/log stability. Figure 5.4 shows the current reliability maps.

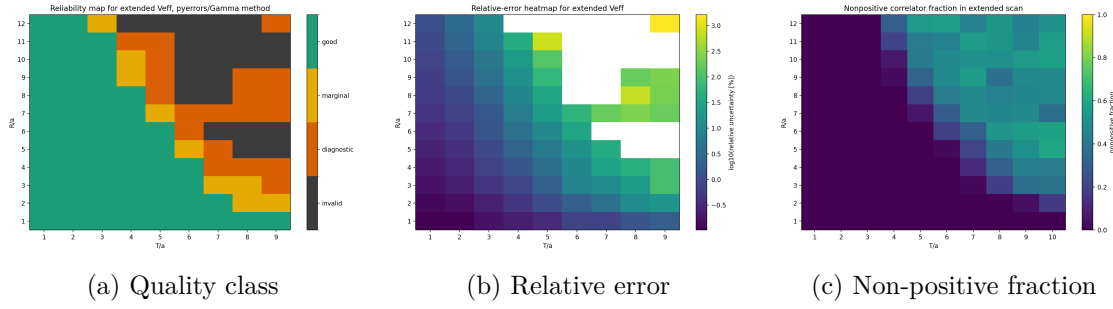


Figure 5.4.: *Preliminary* reliability diagnostics for the extended scan. The maps are used to define safe parameter windows before the flux-tube probe measurements are performed.

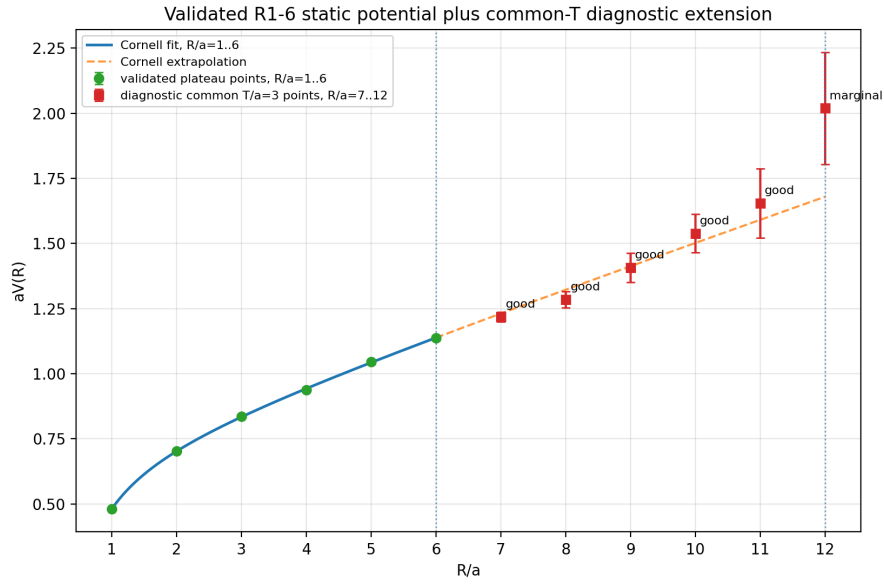


Figure 5.5.: *Preliminary* diagnostic extension of the static potential beyond the validated range using a common $\tau/a = 3$ estimate. The larger- R points are useful for trend checks but are not yet final precision data.

6. Local gluonic probes and flux-tube profiles

6.1. Field-strength estimators

The final physics stage inserts local gluonic probes into the validated Laplace-static source correlator. A plaquette or clover estimator of the field-strength tensor $F_{\mu\nu}$ will be used to form

$$E^2(\mathbf{z}, t) = \sum_{i=1}^3 \text{Tr } F_{i4}(\mathbf{z}, t)^2, \quad (6.1)$$

$$B^2(\mathbf{z}, t) = \sum_{1 \leq i < j \leq 3} \text{Tr } F_{ij}(\mathbf{z}, t)^2, \quad (6.2)$$

and hence the action and energy densities in Eqs. (2.5) and (2.6). The probe should be inserted near the midpoint t_{mid} of the static correlator to maximize ground-state dominance.

6.2. Normalized vacuum-subtracted observable

For the Laplace-static source, the central observable will be

$$\Delta S_{Q\bar{Q}}(\mathbf{z}; R, \tau) = \frac{\langle L(R, \tau) S(\mathbf{z}, t_{\text{mid}}) \rangle}{\langle L(R, \tau) \rangle} - \langle S(\mathbf{z}, t_{\text{mid}}) \rangle. \quad (6.3)$$

An analogous expression will be used for ΔE^2 , ΔB^2 , or $\Delta \epsilon$, depending on which field components have the best signal and the clearest physical interpretation.

6.3. Planned presentation of final results

The final flux-tube chapter should contain:

1. a validation plot showing that the probe insertion does not destabilize the source normalization;
2. transverse profiles at fixed source separation R ;
3. longitudinal profiles along the $Q\bar{Q}$ axis;
4. two-dimensional density maps in the plane containing the sources;
5. systematic comparisons across τ , R , smearing/probe definition, and possibly N_v .

TODO / thesis-writing note

Replace this planning text by the final local-probe measurements once the clover/-plaquette insertion code path is validated.

7. Statistical and systematic uncertainties

7.1. Fit-window dependence

The benchmark static potential relies on identifying plateau regions in $V_{\text{eff}}(R, \tau)$. Fit-window dependence should be reported explicitly by varying the lower and upper time slices, comparing correlated and uncorrelated fits where appropriate, and tracking χ^2/dof and p -values.

7.2. Temporal-source averaging

The present analysis averages over six temporal source positions, $t_0/a = 0, 8, 16, 24, 32, 40$. This reduces statistical noise by using translational symmetry while keeping source positions well separated on the temporal lattice. The thesis should include a comparison between a fixed-source analysis and the temporal-source averaged analysis to demonstrate the effect on uncertainties and central values.

7.3. Autocorrelations

Monte Carlo measurements are generally autocorrelated. The Gamma method estimates statistical errors by explicitly summing autocorrelation functions and choosing an appropriate summation window [4]. For final results, the chosen error-analysis method should be stated consistently for all primary observables, derived observables, and fit parameters.

7.4. Eigenvector truncation and source overlap

The source construction depends on the number of Laplacian eigenvectors and on the eigenmode profile. Current tests use $N_v = 10$ with constant weights. Later production may improve ground-state overlap through larger N_v , Gaussian profiles, or a GEVP basis, following the strategy of optimized Laplace trial states [2].

7.5. Probe definition

For flux-tube profiles, the plaquette or clover definition of $F_{\mu\nu}$, the amount of smearing, and the treatment of vacuum subtraction can all affect the signal. These choices should be separated into physical results, discretization effects, and noise-reduction choices.

8. Conclusions and outlook

The present thesis draft starts from a validated Laplace-static source pipeline. The key benchmark result is that Laplacian-eigenmode static correlators produce stable effective-energy plateaus for $R/a = 1, \dots, 6$, and the resulting potential is described by a Cornell form with acceptable correlated fit quality. This confirms that the source construction is sufficiently mature to normalize local gluonic probe measurements.

The remaining core physics task is to insert local action-density and energy-density probes into the Laplace-static correlator and extract the spatial gluonic response of the gauge field around a static $Q\bar{Q}$ pair. Once this stage is completed, the thesis will connect the static-potential validation with the final flux-tube profiles and discuss the limitations from statistics, operator overlap, probe discretization, and fit-window systematics.

TODO / thesis-writing note

Update this conclusion after final local-probe plots are available. The final version should clearly separate completed results, limitations, and future extensions such as Gaussian profiles, larger N_v , off-axis separations, and hybrid static potentials.

A. Reproducibility notes

A.1. Suggested repository layout

Place this thesis source under the main project repository as:

```
Masterarbeit/  
  laplace_static/  
    qcd/  
      thesis/  
        main.tex  
        tex/  
        frontmatter/  
        chapters/  
        appendices/  
        figures/  
        bib/
```

A.2. Compile commands

A local or Stromboli LaTeX installation can be checked with:

```
command -v latexmk pdflatex biber  
pdflatex --version | head -2  
biber --version
```

Compile from inside `thesis/` with:

```
latexmk -pdf -interaction=nonstopmode main.tex
```

Clean auxiliary files with:

```
latexmk -C
```

A.3. Suggested Git ignore entries

```
# LaTeX build artefacts  
*.aux  
*.bbl  
*.bcf  
*.blg  
*.fdb_latexmk  
*.fls  
*.lof  
*.log  
*.lot  
*.out  
*.run.xml
```

```
*.synctex.gz  
*.toc  
*.xdv  
*.nav  
*.snm  
*.vrb
```

B. Additional diagnostic plots

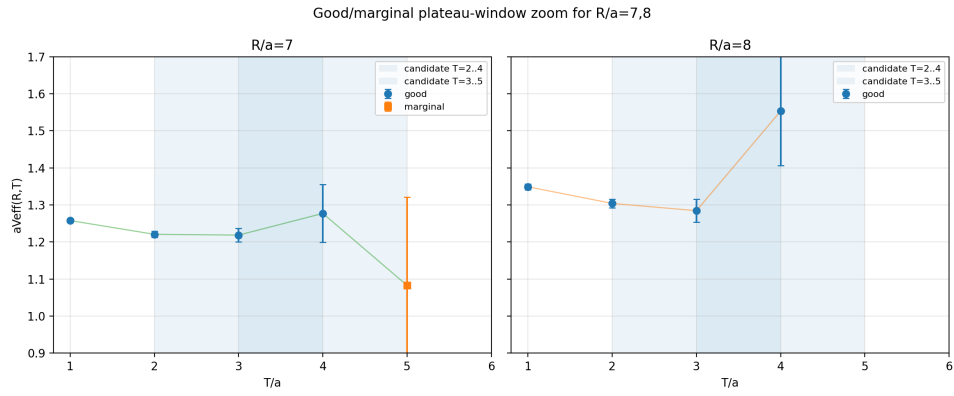


Figure B.1.: *Preliminary* plateau-window comparison for selected larger-distance diagnostic cases.

Bibliography

- [1] Michael Peardon, John Bulava, Justin Foley, Colin Morningstar, Jozef Dudek, Robert G. Edwards, Bálint Joo, Huey-Wen Lin, David G. Richards, and Keisuke Jimmy Juge. “A novel quark-field creation operator construction for hadronic physics in lattice QCD”. In: *Physical Review D* 80 (2009), p. 054506. DOI: 10.1103/PhysRevD.80.054506. arXiv: 0905.2160 [hep-lat].
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